

Chapter 6E

Pituitary Disorders, Diabetes Insipidus, and SIADH

Medvale Internal Medicine - Endocrinology and Metabolism

Coverage note

This chapter uses the uploaded pituitary/endocrine source segment as a topic coverage guide. The prose, algorithms, tables, practice questions, and diagrams are original replacement material for study and graph-RAG indexing.

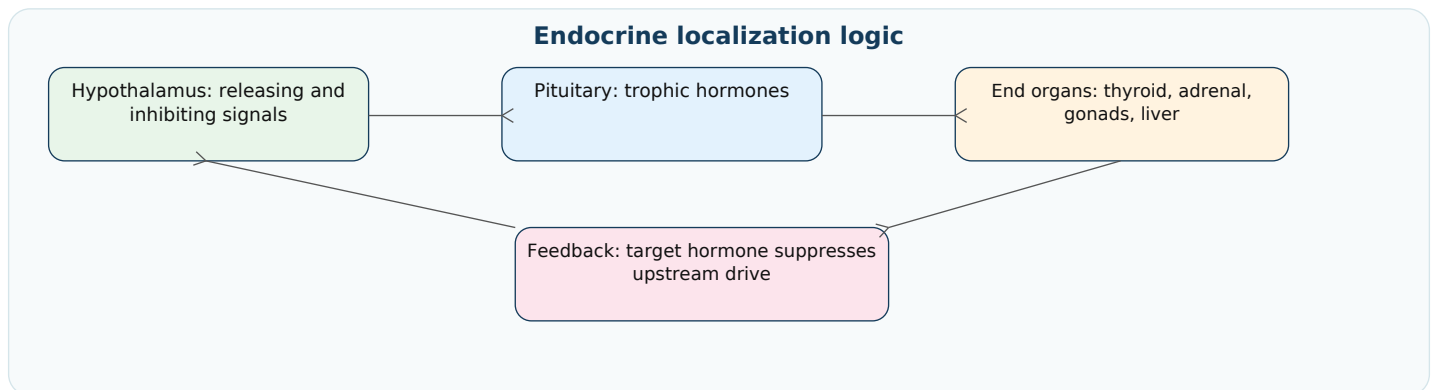
Learning objectives

- Localize pituitary disease by combining the clinical syndrome, target hormone level, and trophic hormone level.
- Distinguish microadenoma from macroadenoma and recognize when visual field loss or hypopituitarism changes urgency.
- Diagnose and manage hyperprolactinemia, acromegaly, hypopituitarism, central diabetes insipidus, nephrogenic diabetes insipidus, primary polydipsia, and SIADH.
- Use urine osmolality, serum osmolality, sodium, volume status, and desmopressin response to classify water balance disorders.
- Connect pituitary disease to reproductive, metabolic, neurologic, renal, and emergency presentations.

Why this matters clinically

Pituitary disease is high-yield because a small lesion can produce a large systemic phenotype. The same sellar mass can present as amenorrhea, erectile dysfunction, headache, bitemporal hemianopsia, unexplained hyponatremia, growth hormone excess, adrenal insufficiency, or diabetes insipidus depending on which cells are secreting, compressed, or destroyed. The clinical question is not simply whether a lesion is present, but whether it is hormonally active, whether it threatens vision, whether it has suppressed normal pituitary function, and whether the patient has an urgent water or adrenal problem.

A pituitary adenoma is classified by size and function. A microadenoma is less than 10 mm. A macroadenoma is at least 10 mm and is more likely to cause headache, optic chiasm compression, hypopituitarism, and stalk effect. Functioning tumors produce a hormone syndrome, such as prolactinoma or growth hormone excess. Nonfunctioning tumors may be clinically silent until they cause mass effect or partial pituitary failure.

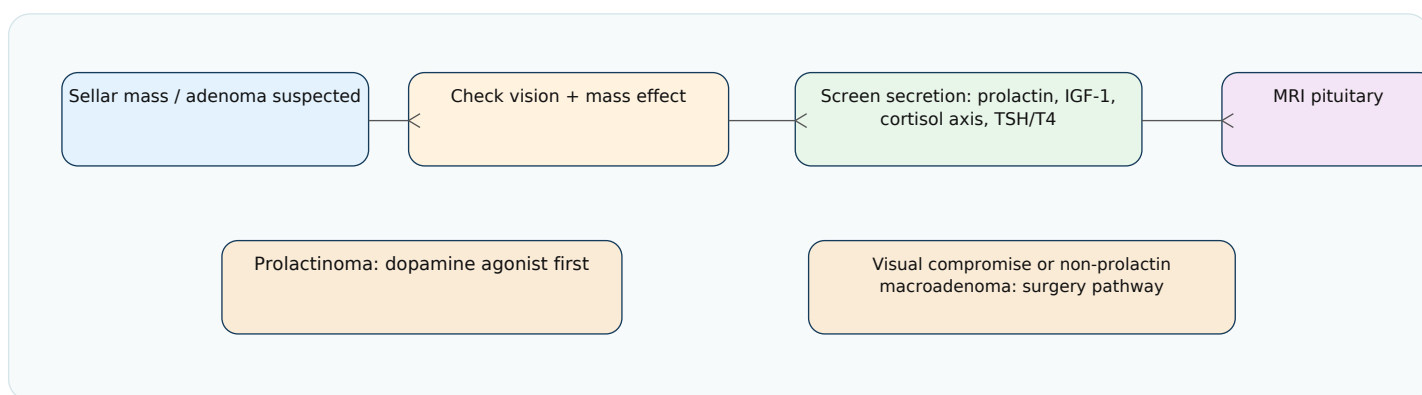


Core illness scripts

Disorder	One-sentence illness script	Anchor tests / clues
Pituitary macroadenoma	Headache, bitemporal visual field loss, and one or more pituitary hormone deficits from a sellar mass.	Pituitary MRI; visual field testing; pituitary hormone panel.
Prolactinoma	Hyperprolactinemia causes hypogonadotropic hypogonadism: amenorrhea, infertility, galactorrhea, low libido, or erectile dysfunction.	Elevated prolactin; pregnancy, hypothyroidism, renal failure, and medications must be excluded.
Acromegaly	GH-secreting adenoma causes soft tissue and skeletal overgrowth plus cardiometabolic disease after epiphyseal closure.	Elevated IGF-1; failure of GH suppression with oral glucose; pituitary MRI.
Hypopituitarism	Loss of anterior pituitary hormones causes adrenal, thyroid, gonadal, growth hormone, or lactotroph deficiency.	Low target hormone with low or inappropriately normal trophic hormone; MRI for structural disease.
Central DI	Low ADH secretion causes large volumes of dilute urine with hypernatremia risk if thirst or water access is impaired.	High/normal serum osmolality, low urine osmolality; urine concentrates after desmopressin.
Nephrogenic DI	Kidney resistance to ADH causes dilute polyuria despite adequate or elevated ADH.	Lithium, hypercalcemia, hypokalemia; little urine concentration after desmopressin.
SIADH	Nonosmotic AVP activity causes hypotonic hyponatremia with inappropriately concentrated urine and clinical euvolemia.	Low serum osm, urine osm >100 mOsm/kg, urine sodium often >30 mmol/L, normal thyroid/adrenal/renal function.

Diagnostic approach to a pituitary mass

A pituitary mass should be evaluated from three angles: mass effect, hormone excess, and hormone deficiency. Mass effect is suggested by headache, visual field loss, ophthalmoplegia, papilledema, or cranial nerve findings. Hormone excess is screened by syndrome-specific testing: prolactin for prolactinoma or stalk effect, IGF-1 for acromegaly, cortisol testing when Cushing disease is suspected, and thyroid testing when TSH-secreting adenoma is possible. Hormone deficiency requires paired target and trophic hormone interpretation; a low free T4 with low or normal TSH is central hypothyroidism, not primary thyroid failure.



Exam trap

Do not treat every enlarged pituitary as a surgical problem. Prolactinomas are usually treated first with dopamine agonists unless there is diagnostic uncertainty, acute vision-threatening compression, medication intolerance, or failure of medical therapy. Conversely, nonfunctioning macroadenomas with visual compromise usually require neurosurgical evaluation.

Hyperprolactinemia

Prolactin secretion is normally restrained by hypothalamic dopamine. Anything that removes dopamine inhibition, increases lactotroph activity, or impairs prolactin clearance can raise the prolactin level. The differential includes prolactinoma, pregnancy, lactation, hypothyroidism, chronic kidney disease, chest wall stimulation, pituitary stalk compression, and medications such as antipsychotics, metoclopramide, verapamil, opioids, and estrogens. Hypothyroidism matters because elevated TRH can stimulate prolactin release; therefore, TSH should be checked before labeling the patient as having a prolactinoma.

The clinical pattern is dominated by suppression of gonadotropin-releasing hormone. Women may develop oligomenorrhea, amenorrhea, infertility, decreased libido, vaginal dryness, or galactorrhea. Men may present later because galactorrhea is uncommon; decreased libido, erectile dysfunction, infertility, gynecomastia, headache, or visual symptoms may be the clue. Chronic hypogonadism can reduce bone mineral density, so hyperprolactinemia is not merely a reproductive complaint.

Step	Reasoning
1. Confirm true elevation	Repeat prolactin if mild elevation and avoid stress or venipuncture artifacts. Very high levels are more consistent with prolactinoma; modest elevations can occur with stalk effect or drugs.
2. Exclude common secondary causes	Pregnancy test, TSH/free T4, renal function, medication review, and clinical context.
3. Image when indicated	Pituitary MRI if persistent unexplained elevation, mass symptoms, or prolactin level strongly suggests adenoma.
4. Treat by cause	Stop offending drug if safe, treat hypothyroidism, or use dopamine agonist therapy for symptomatic prolactinoma. Cabergoline is often preferred because of tolerability and dosing convenience.

Acromegaly

Acromegaly is usually caused by a growth hormone-secreting pituitary adenoma after epiphyseal closure. GH stimulates hepatic and peripheral IGF-1 production; IGF-1 mediates many of the skeletal and soft-tissue findings. The disease often evolves slowly, so patients may not report a single symptom. High-yield clues include increasing ring or shoe size, coarse facial features, frontal bossing, jaw enlargement, macroglossia, carpal tunnel syndrome, oily skin, excessive sweating, arthropathy, obstructive sleep apnea, hypertension, diabetes or impaired glucose tolerance, cardiomyopathy, and colon polyps.

Random GH levels are not reliable because GH secretion is pulsatile. IGF-1 is the best screening test because it integrates GH activity over time. If IGF-1 is elevated or equivocal, an oral glucose suppression test is used: in healthy physiology, glucose suppresses GH; in acromegaly, GH fails to suppress appropriately. Pituitary MRI then localizes the adenoma. Treatment is usually transsphenoidal surgery, followed by somatostatin analogs, GH receptor antagonists, dopamine agonists, or radiotherapy when biochemical disease persists.

Domain	Acromegaly associations
Skeletal/soft tissue	Enlarged hands/feet, coarse facies, jaw growth, macroglossia, arthropathy, carpal tunnel.
Metabolic	Insulin resistance, impaired glucose tolerance, diabetes, dyslipidemia.
Cardiopulmonary	Hypertension, cardiomyopathy, arrhythmia risk, obstructive sleep apnea.
Malignancy/screening	Colon polyps and colorectal cancer risk; colonoscopy strategy depends on age, findings, and local guideline practice.

Hypopituitarism and craniopharyngioma

Hypopituitarism is deficiency of one or more pituitary hormones. Causes include pituitary adenoma, pituitary surgery or radiation, apoplexy, infiltrative disease, traumatic brain injury, Sheehan syndrome, infection, autoimmune hypophysitis, and suprasellar masses. Hormone loss is often partial and sequential. Gonadotropins and growth hormone are commonly affected earlier, while ACTH deficiency is especially dangerous because it can cause adrenal crisis. In central endocrine failure, the target gland hormone is low but the pituitary hormone is low or inappropriately normal rather than appropriately high.

Craniopharyngioma is a suprasellar tumor derived from remnants of Rathke pouch. It may cause headache, bitemporal hemianopsia, papilledema, behavior or cognitive changes, hyperprolactinemia from stalk effect, diabetes insipidus, or panhypopituitarism. Calcification in a suprasellar mass strongly supports the diagnosis. Management often combines neurosurgical resection and radiation depending on size, location, and recurrence risk.

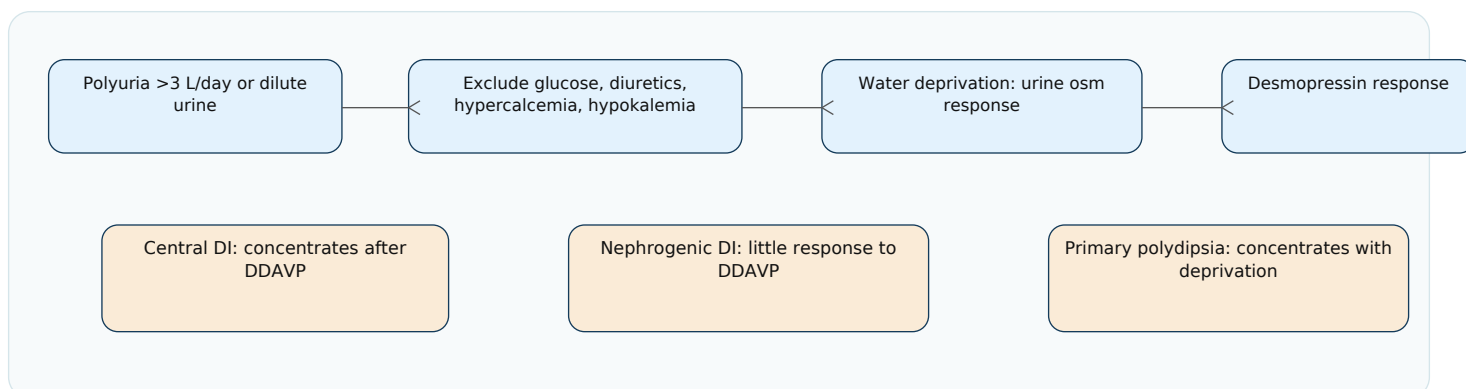
Deficient hormone	Clinical consequence	Lab pattern
ACTH	Secondary adrenal insufficiency: fatigue, hypotension, hyponatremia; hyperkalemia is less typical because aldosterone is largely preserved.	Low cortisol with low/inappropriately normal ACTH.
TSH	Central hypothyroidism: fatigue, cold intolerance, bradycardia, constipation.	Low free T4 with low/inappropriately normal TSH.
LH/FSH	Amenorrhea, infertility, low libido, erectile dysfunction, loss of secondary sex characteristics.	Low estradiol/testosterone with low/inappropriately normal gonadotropins.
GH	Children: growth failure. Adults: increased fat mass, reduced lean mass, low energy, adverse lipid profile.	Low IGF-1 can support deficiency; dynamic testing may be needed.
ADH	Polyuria, polydipsia, hypernatremia risk if posterior pituitary/hypothalamus is involved.	Low urine osmolality relative to plasma osmolality.

Pituitary safety principle

When replacing multiple pituitary hormones, evaluate and treat adrenal insufficiency before starting levothyroxine. Thyroid hormone can increase cortisol clearance and precipitate adrenal crisis in an untreated cortisol-deficient patient.

Diabetes insipidus and the polyuria framework

Diabetes insipidus is a disorder of free-water conservation. Central DI reflects deficient arginine vasopressin/ADH secretion from the posterior pituitary or hypothalamus. Nephrogenic DI reflects renal resistance to ADH. Both cause hypotonic polyuria; the patient may maintain near-normal sodium if thirst and access to water are intact, but hypernatremia can develop quickly when mental status, mobility, perioperative status, or access to water is impaired.



The first step is to confirm true polyuria rather than urinary frequency. A practical adult threshold is more than 3 L/day, though body size and context matter. Next, exclude osmotic diuresis from hyperglycemia, urea, mannitol, or high solute intake. Serum sodium and osmolality help separate water loss from excessive water intake. Primary polydipsia usually produces low or low-normal plasma osmolality and very dilute urine; diabetes insipidus produces inappropriately dilute urine despite normal or high plasma osmolality.

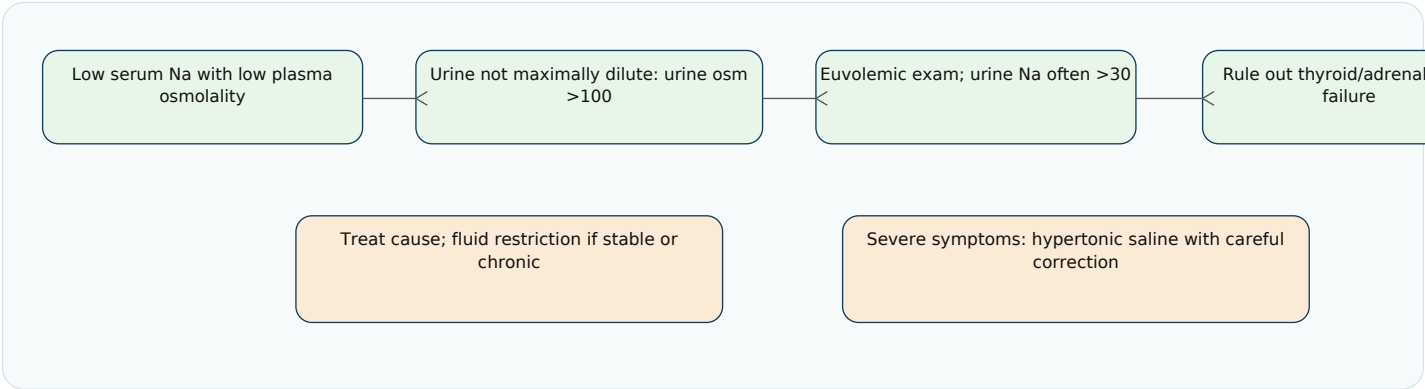
Condition	Serum tendency	Urine before testing	Water deprivation	Desmopressin response
Central DI	High-normal or high Na/osm if water intake cannot keep up	Dilute urine	Urine remains dilute	Urine osmolality rises substantially
Nephrogenic DI	High-normal or high Na/osm if water intake cannot keep up	Dilute urine	Urine remains dilute	Minimal response
Primary polydipsia	Low or low-normal Na/osm	Dilute urine	Urine concentrates with deprivation	Little additional effect after already concentrated

Management of diabetes insipidus

Central DI is treated with desmopressin when persistent and clinically significant. Dosing must be individualized to prevent both dehydration and water intoxication. In hospitalized patients with known cranial DI, missed desmopressin doses, impaired access to water, or perioperative fasting can cause dangerous hyponatremia; medication reconciliation and fluid planning are safety-critical. Nephrogenic DI is managed by treating the cause, stopping lithium when feasible, correcting hypercalcemia or hypokalemia, using dietary sodium restriction, and sometimes using thiazide diuretics or NSAIDs under careful supervision. The paradoxical benefit of thiazides occurs because mild volume contraction increases proximal sodium and water reabsorption, reducing distal water delivery and urine volume.

Syndrome of inappropriate antidiuresis (SIADH)

SIADH is persistent antidiuretic hormone effect despite hypotonicity. Water is retained, serum sodium falls, and urine remains inappropriately concentrated. The patient is usually clinically euvolemic rather than edematous because mild volume expansion suppresses RAAS and increases natriuretic peptide activity, causing sodium excretion. This explains the classic paradox: total body water is increased, but edema is absent and urine sodium is not low.



Common causes include pulmonary disease, CNS disease, malignancy, postoperative pain, nausea, stress, and medications such as SSRIs, carbamazepine, cyclophosphamide, vincristine, and antipsychotics. SIADH is a diagnosis of exclusion. Adrenal insufficiency and hypothyroidism can mimic euvolemic hyponatremia and must be ruled out before the label is secure. Severe or acute hyponatremia can cause headache, nausea, vomiting, confusion, seizures, coma, and respiratory arrest; chronic hyponatremia may be subtler because brain osmolyte adaptation reduces cerebral edema.

Feature	Typical SIADH pattern	Why it matters
Serum sodium	Low	Reflects excess water relative to sodium, not necessarily sodium depletion.
Serum osmolality	Low, usually <275-280 mOsm/kg	Confirms hypotonic hyponatremia rather than pseudohyponatremia or hypertonic hyponatremia.
Urine osmolality	Inappropriately >100 mOsm/kg	Kidney is not maximally diluting urine despite hypotonic plasma.
Urine sodium	Often >30 mmol/L with normal intake	Natriuresis occurs because mild volume expansion suppresses sodium-retaining systems.
Volume status	Clinically euvolemic	No edema; no obvious hypovolemia. Volume assessment is imperfect, so use labs and context.
Exclusions	Normal adrenal, thyroid, renal function; no major diuretic confounder	Prevents misdiagnosis of cortisol deficiency, hypothyroidism, kidney failure, or medication-related hypovolemia.

Treatment of SIADH and hyponatremia safety

Treatment depends on severity, acuity, symptoms, and cause. Stable chronic SIADH is usually managed with treatment of the underlying trigger and fluid restriction. Increasing solute intake, loop diuretics with salt, urea, demeclocycline, or vasopressin receptor antagonists may be used selectively by experienced clinicians. Severe neurologic symptoms require hypertonic saline in a monitored setting. The goal is not to normalize sodium immediately; it is to reverse dangerous cerebral edema while avoiding osmotic demyelination from overcorrection.

Correction safety anchor

A common safety limit is to avoid raising serum sodium by more than about 8 mEq/L in 24 hours in patients at meaningful risk of osmotic demyelination; stricter limits are often used for high-risk patients such as those with alcoholism, malnutrition, advanced liver disease, hypokalemia, or very low starting sodium. Always follow local protocols and check sodium frequently during active correction.

High-yield exam clues and common pitfalls

Clue or pitfall	Interpretation
Bitemporal hemianopsia	Compression of the optic chiasm by a sellar/suprasellar mass until proven otherwise.
Amenorrhea + galactorrhea	Think hyperprolactinemia, but exclude pregnancy and hypothyroidism.
Coarse facial features + enlarged hands + diabetes	Screen for acromegaly with IGF-1; do not use random GH alone.
Polyuria after pituitary surgery	Central DI is a key postoperative concern; monitor urine output, sodium, and osmolality.
Hyponatremia after pituitary disease	Could be SIADH, but also consider secondary adrenal insufficiency and central hypothyroidism.
Euvolemic hyponatremia with concentrated urine	SIADH is likely only after adrenal, thyroid, renal failure, and diuretics are addressed.
Starting levothyroxine in hypopituitarism	Dangerous if cortisol deficiency is untreated; assess adrenal axis first.
Lithium patient with polyuria	Nephrogenic DI until proven otherwise.

Integrated practice questions

Question 1. A 32-year-old woman has 8 months of amenorrhea and milky nipple discharge. Pregnancy test is negative. TSH is markedly elevated and free T4 is low. What is the most important next management step?

Answer/explanation: Treat primary hypothyroidism with levothyroxine. TRH stimulation can raise prolactin; correcting hypothyroidism may resolve the hyperprolactinemia.

Question 2. A 51-year-old man reports larger shoe size, new snoring, carpal tunnel symptoms, hypertension, and intolerance to heat. Which screening test best fits the suspected endocrine disorder?

Answer/explanation: Serum IGF-1. It is the preferred screening test for acromegaly because random GH fluctuates substantially.

Question 3. A patient with traumatic brain injury develops urine output of 9 L/day, serum sodium 153 mEq/L, and serum osmolality 300 mOsm/kg. After desmopressin, urine osmolality rises sharply. What is the diagnosis?

Answer/explanation: Central diabetes insipidus, due to deficient ADH secretion with preserved renal response to desmopressin.

Question 4. A patient on chronic lithium has intense thirst, polyuria, serum sodium 149 mEq/L, and persistently dilute urine with minimal response to desmopressin. What mechanism explains the disorder?

Answer/explanation: Nephrogenic diabetes insipidus. The kidney does not respond adequately to ADH/desmopressin.

Question 5. An elderly patient with small cell lung cancer has sodium 119 mEq/L, low serum osmolality, urine osmolality 900 mOsm/kg, urine sodium 60 mmol/L, no edema, normal creatinine, normal TSH, and normal morning cortisol. What is the most likely diagnosis?

Answer/explanation: SIADH. The urine is inappropriately concentrated in hypotonic hyponatremia, with euvolemia and exclusion of thyroid/adrenal/renal causes.

Question 6. A patient with a pituitary macroadenoma is found to have low free T4 with a normal TSH. Why is this not consistent with primary hypothyroidism?

Answer/explanation: TSH is inappropriately normal. In central hypothyroidism, the pituitary fails to mount the expected TSH response to low thyroid hormone.

Graph-RAG extraction map

The following concept map is designed for downstream node and edge extraction. Nodes should retain aliases and numeric anchors. Edges should preserve directionality and the reason for the connection.

Node	Suggested class	Key facts for node text	High-value edges
Pituitary adenoma	Endocrine neoplasm / sellar mass	Microadenoma <10 mm; macroadenoma ≥10 mm; causes hypersecretion or nonfunctioning.	Causes -> hypersecretion; compresses -> optic chiasm; causes -> hypopituitarism.
Prolactinoma	Pituitary adenoma / reproductive endocrine	Most common functioning pituitary adenoma; dopamine agonists are first-line in many symptomatic patients.	Increases -> prolactin; suppresses -> GnRH; causes -> amenorrhea/infertility/ED/galactorrhea.
Acromegaly	Growth hormone excess	Adult GH excess mediated by IGF-1; screen with IGF-1; confirm with failure of GH suppression after oral glucose.	GH -> IGF-1; causes -> cardiomyopathy/OSA/diabetes; treated by -> transsphenoidal surgery.
Hypopituitarism	Pituitary hormone deficiency	Low target hormone with low or normal trophic hormone; ACTH deficiency is life-threatening.	Caused by -> tumor/surgery/radiation/apoplexy; requires -> hormone replacement.
Central DI	Water balance disorder	ADH deficiency; dilute polyuria; urine concentrates after desmopressin.	Deficiency of -> ADH; causes -> hypernatremia; treated by -> desmopressin.
Nephrogenic DI	Water balance disorder / renal tubular response	ADH resistance; lithium, hypercalcemia, hypokalemia; minimal response to desmopressin	Caused by -> lithium; treated by -> sodium restriction/thiazide; mimics -> central DI.
SIADH	Hyponatremia / ADH excess	Hypotonic hyponatremia with urine osm >100, often urine Na >30, euvolemia, normal adrenal/thyroid/renal function.	Excess ADH -> water retention; causes -> low sodium; treated by -> fluid restriction or hypertonic saline if severe.

Selected references used for clinical cross-checking

- Uploaded endocrine pituitary source segment provided by user: endocrine4.pdf, pages 1-12. Used as a coverage guide only.
- Endocrine Society. Hyperprolactinemia Clinical Practice Guideline resource page, 2011.
- Endocrine Society. Pituitary Incidentaloma Clinical Practice Guideline resource page, 2011.
- Endocrine Society. Acromegaly Clinical Practice Guideline resource page, 2014.
- Society for Endocrinology. Inpatient management of cranial diabetes insipidus guidance, 2018.
- European Society of Endocrinology/European Renal Association. Clinical practice guideline on diagnosis and treatment of hyponatraemia; and related severe symptomatic hyponatraemia emergency guidance.